ScamShield: Evaluation Metrics and Model Details

Hacksploit 2026 · PS 1: Prevention against Digital Arrest and Fraud Calls · Deliverable 5

Generated 29 October 2026 from a fresh run of `EvalHarnessTest` on the current code, including uncommitted changes.

1. Headline result: live-call fraud detection

Test set: 112 labelled call transcripts your team wrote (56 scam, 56 genuine) in English, Hindi and Hinglish. A call counts as caught when it reaches the **red-flag** alert (score of 75 or more).

66.1%	1 / 56	37 / 56	0.133 ms
Accuracy	False positives	False negatives	Latency (median)

Metric	Value	Formula
Accuracy	**66.1%** (74 of 112)	(TP + TN) / n = (19 + 55) / 112
False positives	**1 of 56 genuine (1.8%)**: G-OT-05	FPR = FP / (FP + TN) = 1 / 56
False negatives	**37 of 56 scams (66.1%)**	FNR = FN / (FN + TP) = 37 / 56
Precision	95.0%	TP / (TP + FP) = 19 / 20
Recall	33.9%	TP / (TP + FN) = 19 / 56
Latency	**median 0.133 ms, p95 0.537 ms** (352 runs)	Timed around each scoring call (laptop)
Time to alert	2 spoken lines (median)	First line at which a scam call hit red flag

By split (red flag)

Split	n	TP	FP	TN	FN	Accuracy	Precision	Recall	FPR	FNR
all	112	19	1	55	37	66.1%	95.0%	33.9%	1.8%	66.1%
validation	69	12	1	33	23	65.2%	92.3%	34.3%	2.9%	65.7%
test	43	7	0	22	14	67.4%	100.0%	33.3%	0.0%	66.7%

The warning level (score of 40 or more) gives exactly the same counts: no call in this test stopped at a warning without going on to a red flag.

Recall by scam type

Scam type	n	Caught (red flag)
digital_arrest	16	100.0%
voice_clone_impersonation	3	66.7%
refund_lottery_qr	6	16.7%
kyc_bank_impersonation	8	0.0%
otp_theft	8	0.0%
remote_access	6	0.0%
utility_disconnection	5	0.0%
wrong_number_distraction	4	0.0%

By language (red flag)

Language	n	Accuracy	Recall	FPR	FNR
English	55	61.8%	25.9%	3.6%	74.1%

Language	n	Accuracy	Recall	FPR	FNR
Hindi	26	69.2%	38.5%	0.0%	61.5%
Hinglish	31	71.0%	43.8%	0.0%	56.3%

Supporting models (from app/ml/reports/)

Model	Test data	Accuracy	False positives	False negatives
SMS text (A)	882 real messages	98.3%	6 (0.8%)	9 of 138
Call model (B+C), warning level	6,000 synthetic	87.4%	303 (7.1%)	451 of 1,747
URL / link (D)	25,721 real domains	about 81.6%*	about 2,054 (10.7%)	about 2,671 of 6,435

* URL figures are calculated from the reported rates, because the report does not list exact counts.

2. What runs on a live call

Every detector runs on the phone. There is no cloud model in a call decision. Four steps run in order:

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Caller's speech → (1) phrase matcher → (2) call model → (3) video rules
                → (4) voice rules → score 0–100 → SAFE / WARNING / RED FLAG
```

- Entry point: `app/app/src/main/java/com/scamguard/app/engine/CallRiskScorer.kt`
- Thresholds (`engine/RiskConfig.kt`): **warning at 40 or more, red flag at 75 or more**

Step 1. Phrase matcher (no machine learning)

`collectors/SpeechText.kt` uses hand-written regular-expression lists in English, Hinglish and Hindi (Devanagari). It sorts each clause of speech into 9 categories. Phrases spoken as advice, such as "never share your OTP", are cancelled within the same clause. Voice points are capped at 60 in total.

Category	Example phrases	Points
AUTHORITY_CLAIM	"CBI", "cyber cell", "inspector", "police" (also in Hindi)	15
ACCUSATION	"money laundering", "Aadhaar misused", "FIR registered"	15
ARREST_THREAT	"digital arrest", "warrant", "giraftaar"	25
ISOLATION	"don't disconnect", "phone mat kaatna"	20
PAYMENT_PRESSURE	"transfer to safe RBI account"	15
OTP_REQUEST	"tell me the OTP"	15
CARD_DETAILS	card number, CVV requests	20
REMOTE_APP_REQUEST	"install AnyDesk"	15
LINK_REQUEST	"click the link"	10

Step 2. Call model (trained machine learning, model B+C)

- **Algorithm:** scikit-learn `GradientBoostingClassifier` with 120 trees, max depth 3, learning rate 0.1, subsample 0.8. It predicts 7 classes: LEGIT, OTP_THEFT, MONEY_TRANSFER_PRESSURE, CALL_FORWARDING_HIJACK, CALL_MERGE_FRAUD, REMOTE_ACCESS_FRAUD and DIGITAL_ARREST. It is exported to JSON and run by a pure-Kotlin runtime; tests confirm the Kotlin output matches Python to within 1e-9.
- **Output:** scam probability = 1 − P(LEGIT). Score = probability × 100.
- **Training data: 100% synthetic.** `ml/scamml/callfeat.py` generated 40,000 rows from hand-chosen probabilities (for example, an OTP_THEFT call has a 95% chance of an OTP SMS; a digital-arrest call has a median talk time of 45 minutes). 2% deliberate label noise; split 70/15/15 (28,000 / 6,000 / 6,000).
- **Thresholds** are chosen on the validation split as the lowest threshold reaching a target precision: **warning 0.33** (precision ≥ 0.80) and **red flag 0.73** (precision ≥ 0.95).

- **One-signal guard:** if removing the strongest single factor would drop the probability below the warning threshold, a red flag is lowered to a warning.
- **Most important feature:** talk_minutes, at 40% of importance. That reflects the generator's assumptions, not real scam behaviour.

The 33 input features (all metadata; none come from the audio)

Group	Features
Caller	is_hidden, is_international, number_unavailable, calls_last_hour, reported_number, known_bank_number, user_marked_safe, user_reported_scam
Call	answered, talk_minutes, odd_hour
Money (SMS)	otp_count, otp_untrusted_sender, debit_count, debit_amount_ratio, history_available, first_time_payee, balance_near_zero, otp_then_debit_5min
Device	forwarding_enabled, second_call_merged, remote_app_opened, sideloaded_install, link_received, link_then_install, network_lost
Other models	sms_analyzed, max_sms_scam_prob, sms_impersonation, sms_credential, sms_payment, sms_remote_access, max_url_prob

Synthetic test results (n = 6,000; 1,747 scams)

Threshold	Precision	Recall	F1	FPR	ROC-AUC	TN / FP / FN / TP
Warning 0.33	0.811	0.742	0.775	0.071	0.906	3950 / 303 / 451 / 1296
Red flag 0.73	0.955	0.598	0.736	0.012	0.906	—

Scam-type accuracy on scams: 0.528.

Steps 3 and 4. Video rules and voice rules (fixed rules on top)

`DigitalArrestRules.kt` and `VoiceRules.kt` add points, then raise the score to a minimum when certain combinations of stages appear in the same call. **A red flag from voice evidence needs one of these combinations;** points alone cannot produce one.

Combination heard in one call	Minimum score
authority + accusation	40 (warning)
(authority or accusation) + (arrest or isolation), or arrest + payment	75 (red flag)
credential request (OTP, card or remote app) + (authority or payment)	75 (red flag)
authority/accusation + threat + (money or codes)	90

Video calls also get points for an unknown video caller (15), a long call (20 at 20 min, 35 at 60 min), an authority-style caller name (25) and on-screen text (law enforcement 30, legal threat 25, payment demand 15; capped at 50).

3. Models used outside the call score

	SMS model (A)	Link model (D)
Algorithm	TF-IDF (word 1–2-grams + character 3–5-grams, sublinear, L2) → logistic regression (C = 16); multinomial category head	TF-IDF over character 3–5-grams of the registered domain, plus TLD, length, hyphen and digit buckets → logistic regression (C = 1)
Data	**Real:** UCI SMS Spam + Mendeley SMS Phishing. 11,545 rows, 5,876 after removing near-duplicates (915 malicious). 1,647 synthetic Indian scam messages used for training only.	**Real:** PhiUSIIL. 235,795 URLs → 171,478 unique domains (42,904 malicious).
Split	70/15/15 stratified, grouped so near-duplicate messages never land in both train and test (train 4,113 / val 881 / test 882)	Grouped by domain, so no domain is in both train and test (train 120,035 / val 25,722 / test 25,721)
Threshold	0.394 (best F1 on validation)	0.328 "uncertain"; 0.805 "likely phishing"

	SMS model (A)	Link model (D)
Test result	TN 738, FP 6, FN 9, TP 129. **Accuracy 98.3%**, precision 0.956, recall 0.935, F1 0.945, FPR 0.8%, ROC-AUC 0.995	At 0.328: precision 0.647, recall 0.585, F1 0.614, **FPR 10.7%**, ROC-AUC 0.829. At 0.805: precision 0.976, recall 0.240
Notes	Messages with fewer than 2 tokens or under 40% known words get "cannot judge". Synthetic holdout F1 0.911 (not real-world).	Weak by design: it sees only the registered domain, so it cannot catch phishing on compromised legitimate domains or in URL paths.

Both models feed the call model through its `max_sms_scam_prob` and `max_url_prob` features.

The website (`website/src/lib/semantic/`) has its own layer: a multilingual MiniLM sentence-embedding similarity check (Xenova/paraphrase-multilingual-MiniLM-L12-v2) and an optional Anthropic or Gemini LLM verdict. **It is not part of any metric in this document.**

4. How the headline metrics are calculated

Code: `app/app/src/test/java/com/scamguard/app/eval/EvalHarnessTest.kt`

- **Data:** `app/src/test/resources/eval/transcripts.jsonl` holds 112 transcripts (56 scam, 56 genuine), each labelled scam or genuine with a type, language and split (69 validation, 43 test).
- **Replay:** each transcript is played as an unknown incoming call. The phone rings, the call is answered, then each line arrives 30 seconds apart. Every line passes through `SpeechText` and then the real `CallRiskScorer`, using the shipped models with the voice layer on. Nothing is tuned on this data.
- **Counting:** a call counts as flagged if it **ever** reached the red-flag level during the call, because the app alerts the first time that happens. A scam call is the positive class.
- **Formulas** (`Confusion` class): accuracy = (TP+TN)/n; precision = TP/(TP+FP); recall = TP/(TP+FN); FPR = FP/(FP+TN); FNR = FN/(FN+TP).
- **Latency:** `System.nanoTime()` is recorded around each `scorer.evaluate(...)` call, giving 352 timings after 3 warm-up calls so Java start-up is not counted. The percentile is the value at index $(n-1) \times q$ of the sorted list: median 0.133 ms, p95 0.537 ms, max 0.784 ms. This is text-to-decision time on a laptop only; speech-to-text on the phone is not included (the app measures that under Settings → Evaluation and metrics).

Reproduce: `cd app && ./gradlew :app:testDebugUnitTest --tests '*EvalHarnessTest*'`

5. Why recall is only 34%

- Every eval call is an unknown caller with no SMS, no debit, no forwarding and short talk time. **The call model cannot hear speech**, so it gives every call in this test a low base probability.
- A call reaches a red flag only when one of the voice combinations fires. Digital-arrest scripts trigger them (100% caught). OTP-theft scripts ("tell me the OTP") mostly trigger a single category worth 15 points and end at a score of 14–35, **below even the warning level of 40**.
- This is also why the warning and red-flag columns are identical.

Errors

- **Missed scams (37):** all 8 otp_theft, all 8 kyc_bank_impersonation, all 6 remote_access, 5 of 6 refund_lottery_qr, all 5 utility_disconnection, all 4 wrong_number_distraction, and S-VC-03 (voice_clone_impersonation). Peak scores 14–35.
- **False alarm (1):** G-OT-05 (genuine, type "other"), peak score 75.

6. Before you present this

Limits. The call model's own report says its synthetic scores (87% accuracy, ROC-AUC 0.91) only show it fits its generator; they are **not real-world accuracy**. The 112 transcripts were written by your team and are text only, so they include no speech-to-text errors, and each call is tested without SMS, reputation or app signals. The SMS model's 98.3% is the only figure measured on real held-out data. Real-call accuracy will differ.